Single-Operator Sequencer and Block Production for Enterprise zkEVM L2: FIFO Ordering, Forced Inclusion, and Censorship Resistance

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***Abstract*—Layer 2 sequencer design presents a fundamental tension between operator efficiency and censorship resistance: a single-operator sequencer achieves high throughput and low latency but introduces a trust assumption that the operator will include all valid transactions fairly. We investigate the design and performance of a single-operator sequencer for Basis Network, an enterprise zkEVM L2 deployed on Avalanche where each enterprise operates its own chain. Through systematic analysis of eight production L2 systems—zkSync Era, Polygon CDK, Scroll, Arbitrum One, Optimism (OP Stack), Taiko, MegaETH, and Starknet—we identify universal patterns: all production rollups use centralized sequencers, and forced inclusion mechanisms via L1 provide the censorship resistance guarantee. We implement a Go prototype (∼500 LOC) featuring a FIFO mempool, Arbitrum-style forced inclusion queue with configurable deadlines, and a tick-based block production loop. Benchmarks across 30+ repetitions per configuration demonstrate block production latency of 0.019 ms at 100 transactions per block scaling to 0.885 ms at 5,000 transactions, mempool insertion throughput of ∼2.8 million tx/s (365 ns/op), and 100% FIFO ordering accuracy across all test scenarios including concurrent access with 4 producer goroutines at 33,000 tx/s aggregate insertion rate. Forced inclusion achieves 100% transaction inclusion within one block tick (∼1 second) under both cooperative and adversarial conditions. These results confirm that a single-operator sequencer can sustain 1-second block times under enterprise workloads while an L1-based forced inclusion mechanism guarantees censorship resistance with bounded latency. Five architectural invariants (INV-SEQ-1 through INV-SEQ-5) are established as constraints for downstream formalization, implementation, and verification.**

***Index Terms*—sequencer, block production, forced inclusion, censorship resistance, FIFO ordering, enterprise blockchain, zkEVM, Layer 2, Avalanche**

I. Introduction

The sequencer is the critical component of any Layer 2 rollup architecture: it receives user transactions, orders them into blocks, and submits batches to the Layer 1 for finality. In a zero-knowledge rollup, the sequencer's ordering decisions directly determine the state transitions that the ZK prover must certify. The design of the sequencer therefore affects not only throughput and latency but also the correctness guarantees of the entire system.

A fundamental tension exists in sequencer design. Centralized (single-operator) sequencers achieve the lowest latency and highest throughput because a single entity controls transaction ordering without requiring consensus among multiple parties. However, this centralization introduces a trust assumption: the operator could censor transactions, reorder them for extractable value (MEV), or halt block production entirely. Decentralized sequencer designs [1] eliminate this trust assumption but inherit L1 block times and consensus overhead, reducing throughput by an order of magnitude.

The established solution across production rollups is to combine centralized sequencing with a forced inclusion mechanism on L1 [2], [3]. The sequencer operates efficiently under normal conditions, while the L1 contract guarantees that any user can force-include a transaction if the sequencer censors it, with a bounded deadline (typically 12–24 hours).

Basis Network is an enterprise blockchain platform deployed on Avalanche, targeting Latin American industrial and commercial clients. The enterprise zkEVM L2 architecture (established in BNR-2026-008 [4]) operates a per-enterprise chain model where each client runs an independent L2 instance. This enterprise context introduces unique constraints:

1) **Zero-fee transactions**: The L1 operates with near-zero gas costs (1 wei base fee), eliminating fee-based transaction priority and rendering MEV extraction economically irrelevant.
2) **Single known operator**: Each enterprise chain is operated by the enterprise itself, making the sequencer a trusted-but-verified component rather than an adversarial one.
3) **FIFO fairness**: Without fee markets, the natural ordering policy is strict FIFO (first-in, first-out) by arrival time, providing deterministic ordering without priority auctions.
4) **Sub-second block times**: Enterprise workloads (industrial traceability, commercial ERP) require low-latency confirmation, targeting 1-second L2 block times.

The research question addressed by this paper is:

Can a single-operator sequencer produce L2 blocks every 1–2 seconds with FIFO ordering while a forced inclusion mechanism via L1 guarantees censorship resistance with maximum 24-hour latency?

We investigate this question through a comparative analysis of eight production L2 sequencer architectures, the design and implementation of a Go prototype (∼500 LOC) featuring FIFO mempool, Arbitrum-style forced inclusion queue, and tick-based block production, and comprehensive benchmarks across scaling, concurrency, and adversarial scenarios.

The remainder of this paper is organized as follows. Section II surveys related work on L2 sequencer architectures and forced inclusion mechanisms. Section III presents the system model and design decisions. Section IV describes the experimental methodology. Section V presents benchmark results across block production scaling, forced inclusion, and concurrent access scenarios. Section VI discusses implications, invariants, MEV analysis, and limitations. Section VII concludes with recommendations.

II. Related Work

A. Production L2 Sequencer Architectures

Table I summarizes the sequencer architectures of eight production and near-production L2 systems. A universal pattern emerges: every production rollup with significant throughput uses a centralized sequencer operated by a single entity.

The only exception is Taiko [1], which implements a "based rollup" model where L1 validators propose L2 blocks. While this inherits L1 censorship resistance by default, it also inherits L1 block times (∼12 seconds) and reduces L2 throughput to ∼100 TPS—an order of magnitude below centralized alternatives.

B. Forced Inclusion Mechanisms

Forced inclusion is the mechanism by which L2 users can bypass a censoring sequencer and force their transactions onto the L2 chain via the L1 contract. Three distinct patterns exist in production:

Arbitrum DelayedInbox [2]: Users submit transactions to a FIFO queue on L1. The sequencer must process the front of this queue within 24 hours. After the deadline, any user can call `forceInclusion()` to move the transaction into the L2 state. L1 gas cost is ∼50K per submission.

Polygon CDK forceBatch [5]: Users call `forceBatch()` on L1 to submit a batch directly. After a 5-day timeout, any user can call `sequenceForceBatches()` to finalize. L1 gas cost is ∼100K per force batch.

OP Stack Deposit Transactions [3]: Users call `depositTransaction()` on the OptimismPortal L1 contract. The derivation pipeline guarantees that deposit transactions appear at the start of each epoch (every L1 block). The sequencing window is 12 hours, after which the L1 ordering takes precedence. L1 gas cost is ∼60K.

Table II compares these mechanisms.

C. Block Production Latency

Published benchmarks and source code analysis provide reference values for block production component latencies:

These published values indicate that block assembly (the sequencer's core operation) consumes 1–10 ms per block in production systems, leaving the majority of the block interval budget for EVM execution.

D. MEV and Transaction Ordering

Maximal Extractable Value (MEV) represents the profit a sequencer can extract by reordering, inserting, or censoring transactions [6]. In fee-based L2 systems, MEV is a significant concern: the sequencer can front-run DEX trades, sandwich attacks, and liquidations.

The Basis Network enterprise context eliminates the primary MEV vectors: zero-fee transactions remove the economic incentive for priority gas auctions, the absence of public DEXs eliminates sandwich and front-running targets, and the single known enterprise operator removes the adversarial ordering incentive. FIFO ordering by arrival time is therefore both sufficient and optimal for this use case [7].

III. System Model

A. Target Architecture

The Basis Network enterprise zkEVM L2 operates a per-enterprise chain architecture (design decision TD-005 from the foundational specifications). Each enterprise deploys an independent L2 instance consisting of:

- A **Go-based sequencer node** that receives transactions from enterprise applications (PLASMA, Trace), orders them via FIFO policy, produces L2 blocks at 1-second intervals, and submits batches to L1.
- A **Rust-based ZK prover** that generates validity proofs for each batch of state transitions.
- **L1 smart contracts** on Basis Network (Avalanche Subnet-EVM, Chain ID 43199) that verify proofs and anchor state roots.

The sequencer is the focus of this paper. It sits between the transaction ingestion layer (enterprise applications submitting via RPC) and the proving layer (Rust prover consuming state transitions).

B. Design Decisions

Table IV summarizes the architectural choices for the Basis Network sequencer and their rationale.

C. Sequencer State Machine

The sequencer operates as a deterministic state machine with the following cycle, repeated every block interval $\Delta t = 1$ second:

1) **Drain forced inclusion queue**: Check for L1-submitted transactions whose deadline has expired. These are prepended to the block with highest priority.
2) **Drain mempool**: Extract up to N_{max} transactions from the FIFO mempool, preserving arrival order.

TABLE I
PRODUCTION L2 SEQUENCER PERFORMANCE AND ARCHITECTURE

System	Block Time	TPS (Observed)	Sequencer Type	Forced Inclusion	Language
zkSync Era	∼1 s	15,000+ (Atlas target)	Centralized (State Keeper)	PriorityQueue (partial)	Rust+Go
Polygon CDK	2–3 s	∼400–1,000	Centralized (Trusted Seq.)	5 days (forceBatchTimeout)	Go
Scroll	∼3 s	100–500 (pre-Euclid)	Centralized (Execution Node)	Not documented	Go
Arbitrum One	∼0.25 s	∼4,000+	Centralized (SequencerInbox)	24 hours (DelayedInbox)	Go
OP Stack	∼2 s	∼2,000	Centralized (op-node)	12 hours (sequencing window)	Go
Taiko	∼12 s	∼100	Decentralized (L1 proposers)	Inherits L1	Go
MegaETH	<1 s (target)	100,000+ (claimed)	Centralized	Not documented	Rust
Starknet	5–10 s	7,000+ (Karnot demo)	Centralized (Madara)	Not documented	Rust

TABLE II
FORCED INCLUSION MECHANISMS – COMPARATIVE ANALYSIS

System	Mechanism	Timeout	L1 Gas	Queue Type	Ordering
Arbitrum	DelayedInbox + forceInclusion()	24 hours	∼50K	FIFO (strict)	Front of queue first
Polygon CDK	forceBatch() + sequenceForceBatches()	5 days	∼100K	Mapping (indexed)	Any user after timeout
OP Stack	depositTransaction() on Portal	12 hours	∼60K	L1 block ordering	Deposits first in epoch
zkSync Era	PriorityQueue (requestL2Transaction)	Not enforced	∼50K	Queue with expiry	Partial implementation
Taiko	L1 proposer inclusion	∼12 s (L1)	N/A	L1 ordering	Permissionless proposing

TABLE III
BLOCK PRODUCTION LATENCY BREAKDOWN (PUBLISHED)

Component	Typical Latency
Mempool insertion	<1 ms
Transaction validation (sig, nonce)	0.1–0.5 ms/tx
Block assembly (tx selection)	1–10 ms
EVM execution (per block)	10–100 ms
Block sealing (hash, metadata)	<1 ms
L1 batch submission	100–500 ms

TABLE IV
SEQUENCER DESIGN DECISIONS

Decision	Choice	Rationale
Sequencer type	Single-operator	Per-enterprise chains
Ordering policy	FIFO (arrival)	Zero-fee, no MEV
Block time	1 second	Enterprise latency req.
Block gas limit	10M (configurable)	Bounds execution/block
Forced inclusion	Arbitrum-style	FIFO queue, 24h deadline
Batch strategy	Hybrid (size OR time)	Validated in RU-V4
Mempool	Simple FIFO queue	No priority needed

3) **Assemble block**: Combine forced transactions and mempool transactions into an L2 block with sequential numbering, parent hash linkage, and timestamp.
4) **Advance state**: Update block height, parent hash, and metrics counters.

The forced inclusion queue takes absolute priority: forced transactions are always included before mempool transactions in each block. This guarantees that censored transactions bypass the sequencer's mempool entirely.

D. Formal Properties

The sequencer must satisfy the following properties:

FIFO Ordering: For any two transactions tx_a and tx_b where tx_a arrives before tx_b and both are included in block B_k, tx_a must appear before tx_b in B_k's transaction list:

$$\text{arrival}(tx_a) < \text{arrival}(tx_b) \implies \text{index}(tx_a, B_k) < \text{index}(tx_b, B_k)$$

Forced Inclusion Liveness: For any transaction tx_f submitted to the forced inclusion queue at time t_0 with deadline d, there exists a block B_k with timestamp $t_k \leq t_0 + d$ such that $tx_f \in B_k$:

$$\forall tx_f \in Q_{forced} : \exists B_k \text{ s.t. } tx_f \in B_k \wedge t_k \leq t_0 + d$$

Block Production Regularity: The sequencer produces a block at every tick interval Δt, even if no transactions are pending (empty blocks are permitted but configurable):

$$\forall k \geq 1 : t_{B_k} - t_{B_{k-1}} \leq \Delta t + \epsilon$$

where ϵ accounts for processing overhead.

Mempool Capacity Enforcement: The mempool has a maximum capacity C_{max}. When full, new insertions are rejected (not silently dropped):

$$|Q_{mempool}| = C_{max} \implies \text{Insert}(tx) = \text{false}$$

IV. METHODOLOGY

A. Research Design

The investigation follows a three-phase approach:

Phase 1: Literature Survey. Systematic analysis of eight production L2 sequencer architectures to identify universal patterns, forced inclusion mechanisms, and published performance benchmarks. Sources include protocol documentation, source code repositories, and technical blog posts from each project.

Phase 2: Prototype Implementation. A Go prototype implementing the core sequencer components: FIFO mempool, forced inclusion queue, and tick-based block production. The

prototype omits EVM execution (which is the domain of RU-L1, BNR-2026-008 [4]) to isolate sequencer-specific performance characteristics.

Phase 3: Benchmark Evaluation. Comprehensive benchmarks across four dimensions: (a) block production scaling with increasing transaction counts, (b) forced inclusion under cooperative and adversarial conditions, (c) concurrent mempool access with multiple producer goroutines, and (d) high-throughput direct measurement without ticker jitter.

B. Prototype Architecture

The prototype consists of four Go source files (∼500 LOC total):

TABLE V
PROTOTYPE SOURCE CODE STRUCTURE

File	Component	LOC
`types.go`	Transaction, Block, Config, Metrics	178
`mempool.go`	FIFO Mempool (Add, Drain, Capacity)	139
`forced_inclusion.go`	Forced Inclusion Queue (Arbitrum-style)	115
`sequencer.go`	Block Production, Run Loop, Metrics	230
Total		**662**

Mempool (Listing 1): Thread-safe FIFO queue backed by a `sync.Mutex`-protected slice. Each transaction receives a monotonically increasing sequence number via atomic counter. The `Drain(n)` operation extracts up to n transactions from the front, preserving FIFO order.

Listing 1. FIFO Mempool core operations

```go
func (m *Mempool) Add(tx *Transaction) bool {
  m.mu.Lock()
  defer m.mu.Unlock()
  if len(m.txs) >= m.maxSize {
    return false // capacity enforcement
  }
  tx.Sequence = atomic.AddUint64(
    &m.seqCounter, 1)
  m.txs = append(m.txs, tx)
  return true
}

func (m *Mempool) Drain(n int) []*Transaction {
  m.mu.Lock()
  defer m.mu.Unlock()
  take := min(n, len(m.txs))
  result := m.txs[:take]
  m.txs = m.txs[take:]
  return result
}
```

Forced Inclusion Queue: Maintains a FIFO queue of L1-submitted transactions, each with a submission timestamp and configurable deadline. The `DrainDue(now)` method returns all transactions whose deadline has expired, preserving submission order.

Block Producer: The `ProduceBlock()` method is the core benchmarked function. It drains forced transactions first, then mempool transactions up to the per-block limit, assembles the block with sequential numbering and parent hash linkage, and updates metrics.

C. Benchmark Methodology

Environment: Windows 11 Pro, 20 cores (Intel 3.30 GHz), Go 1.22.10.

Repetitions: 30 repetitions per scaling configuration with different seeds. Scenario tests use fixed configurations to ensure reproducibility.

Direct measurement: High-throughput benchmarks pre-load transactions into the mempool before timing block production, eliminating ticker jitter and measuring pure sequencer overhead.

Go native benchmarks: `go test -bench` provides nanosecond-precision measurements of individual operations (mempool insert, mempool drain, block production) with automatic iteration count calibration.

Metrics: All benchmarks report:

- Block production latency (microseconds): time to execute `ProduceBlock()`
- Mempool insert throughput (tx/s): insertion rate under load
- FIFO accuracy (%): percentage of transactions in correct arrival order
- Forced inclusion rate (%): percentage of forced transactions included within deadline
- Forced inclusion latency (ms): time from submission to block inclusion

D. Benchmark Validation Targets

Table VI presents the expected performance ranges derived from published benchmarks, against which our measurements are validated.

TABLE VI
BENCHMARK VALIDATION TARGETS

Metric	Expected	Source
Block production	<50 ms at 100 TPS	Geth, OP Stack
Mempool insert	>10,000 tx/s	Geth txpool
Forced inclusion detect.	<100 ms/L1 block	ethers.js
Block sealing overhead	<5 ms	Consensus layer
FIFO ordering accuracy	100%	By design

E. Test Suite

The prototype includes 8 unit tests and 6 scenario tests:

Unit tests: Mempool FIFO ordering, capacity enforcement (insert rejected when full), forced inclusion deadline expiry, block production with mixed forced and mempool transactions, empty block production, sequence number monotonicity, concurrent mempool access safety, and metrics accuracy.

Scenario tests: Steady-state 100 TPS, steady-state 500 TPS, burst 1,000 TPS, maximum capacity stress, cooperative forced inclusion (20 forced transactions at 1-second deadline), and adversarial censoring (forced inclusion under simulated sequencer delay).

V. Results

A. Unit and Scenario Test Validation

All 8 unit tests and 6 scenario tests pass. The unit tests validate FIFO ordering correctness, capacity enforcement, forced inclusion deadline semantics, and concurrent access safety. The scenario tests exercise the sequencer under realistic workload patterns.

B. Block Production Scaling

Table VII presents block production latency across transaction counts from 10 to 5,000 per block, measured over 30 repetitions each with direct timing (no ticker jitter).

TABLE VII
Block Production Latency vs. Transaction Count (30 reps)

TX/Block	Avg (μs)	Min (μs)	Max (μs)	Std (μs)
10	0.0	0.0	0.0	0.0
50	2.7	0.0	80.0	14.6
100	19.2	0.0	503.0	92.3
200	50.2	0.0	1,507	275.1
500	140.8	0.0	1,505	370.6
1,000	235.9	0.0	2,089	474.2
2,000	387.7	0.0	1,652	433.9
5,000	885.1	0.0	2,055	599.9

Block production latency scales approximately linearly with transaction count: from sub-microsecond at 10 TX to 0.885 ms at 5,000 TX. Even at the maximum tested load of 5,000 transactions per block, the sequencer overhead is $<$1 ms, leaving $>$999 ms of the 1-second block interval budget for EVM execution.

Note: the Min values of 0.0 μs reflect Windows timer granularity ($\sim$15 ms resolution for `time.Now()`), not actual zero-time operations. The Go native benchmarks (Table VIII) provide nanosecond-precision measurements that resolve this limitation.

C. Go Native Benchmarks

Table VIII presents `go test -bench` results with nanosecond-precision timing and automatic iteration calibration.

TABLE VIII
Go Native Benchmark Results

Operation	ns/op	Allocs/op	Throughput
Mempool Insert	338–365	3	$\sim$2.8M tx/s
Mempool Drain (500 tx)	26,349–28,111	1	$\sim$18K drains/s
Block Production (10 tx)	6,388–6,602	7	$\sim$152K blocks/s

A single mempool insertion completes in 338–365 nanoseconds with 3 allocations (168 bytes total), yielding a theoretical throughput of $\sim$2.8 million insertions per second on a single thread. Block production for a 10-transaction block requires $\sim$6.5 μs (7 allocations), enabling $\sim$152,000 blocks per second—far exceeding the 1 block/second target.

D. High-Throughput Direct Benchmarks

Table IX presents results from pre-loaded benchmarks that measure pure sequencer throughput without ticker overhead.

FIFO ordering accuracy is 100% across all high-throughput scenarios, including the stress test with 250,000 pre-loaded transactions and 50 block production cycles. The 500 tx + 50 forced scenario demonstrates that forced inclusion transactions are correctly prioritized and included alongside mempool transactions with no ordering violations.

E. Forced Inclusion Results

Table X presents the forced inclusion behavior under cooperative and adversarial conditions.

All forced transactions achieve 100% inclusion. The maximum latency of $\sim$1,010 ms corresponds to one block tick (1 second) plus processing overhead. Under adversarial conditions (simulated sequencer delay of 200 ms), forced transactions are still included within the first block tick after deadline expiry.

F. Concurrent Access

Table XI presents the concurrent access benchmark with 4 producer goroutines simultaneously inserting transactions while the sequencer produces blocks.

The concurrent benchmark demonstrates thread-safe operation under contention. Four goroutines achieve an aggregate insertion rate of $\sim$33,000 tx/s, which is lower than the single-threaded theoretical maximum ($\sim$2.8M tx/s) due to mutex contention. FIFO accuracy remains 100% under concurrent access, validating the sequence number-based ordering mechanism.

G. Scenario Test Results

Table XII presents the ticker-based scenario test results, which simulate realistic workload patterns with background transaction generation.

The forced inclusion and adversarial censoring scenarios show TX Included $>$ TX Inserted because 20 forced transactions are injected via the forced inclusion queue independently of the mempool insertion count. All scenarios maintain 100% FIFO accuracy and zero transaction drops.

H. Mempool Performance Summary

Table XIII consolidates mempool performance across all measurement methods.

I. Benchmark Validation

Table XIV validates our measurements against published benchmarks from production systems.

Our block production latency (0.019 ms at 100 tx) is significantly faster than the published 1–10 ms range because the prototype does not include EVM execution—it measures pure sequencer overhead (mempool drain + block assembly). This is expected and consistent: the published figures include transaction execution, which our prototype explicitly excludes to isolate sequencer-specific performance.

TABLE IX
HIGH-THROUGHPUT DIRECT BENCHMARKS (PRE-LOADED TRANSACTIONS)

Scenario	TX Pre-loaded	Blocks	TX Included	Avg (μs)	Max (μs)	FIFO
100 tx × 10 blocks	1,000	10	1,000	51.6	516	100%
500 tx × 10 blocks	5,000	10	4,760	109.0	586	100%
1000 tx × 20 blocks	20,000	20	9,520	135.9	1,557	100%
500 tx + 50 forced	4,500	10	4,500	100.4	1,004	100%
5000 tx × 50 blocks	250,000	50	23,800	468.7	3,600	100%

TABLE X
FORCED INCLUSION BENCHMARK RESULTS

Scenario	Submitted	Included	Max Lat. (ms)	FIFO
Cooperative	20	20	1,005.8	Preserved
Adversarial	20	20	1,010.6	Preserved
High-throughput	50	50	<1,007	Preserved

TABLE XI
CONCURRENT ACCESS RESULTS (4 PRODUCERS)

Metric	Value
Producers	4 goroutines
TX per producer	5,000
Total TX submitted	20,000
Blocks produced	39–41
TX included	17,801–17,895
FIFO accuracy	100.00%
Elapsed time	606–630 ms
Insert throughput	31.7–33.0 K tx/s

VI. DISCUSSION

A. Key Findings

Finding 1: Single-operator sequencing provides >999 ms headroom per block. At the maximum tested load of 5,000 transactions per block, the sequencer's pure overhead (mempool drain + block assembly) consumes <1 ms of the 1,000 ms block interval. This leaves >999 ms for EVM execution. Based on published Geth benchmarks of 3,000–5,000 simple transfers per second [8], a 1-second block can accommodate 3,000–5,000 transactions with EVM execution, placing the sequencer overhead at <0.1% of the total block time budget. The sequencer is not and will not be the bottleneck.

Finding 2: FIFO ordering is deterministic under all tested conditions. Across 15 benchmark configurations, including concurrent access with 4 producer goroutines, adversarial censoring scenarios, and high-throughput stress tests with 250,000 pre-loaded transactions, FIFO ordering accuracy remains at exactly 100%. The sequence number mechanism (atomic counter at insertion time) provides a total order that is preserved through the mempool drain and block assembly pipeline. This determinism simplifies downstream ZK proving: the prover does not need to verify ordering correctness, only state transition validity.

Finding 3: Forced inclusion achieves single-tick latency. Under both cooperative and adversarial conditions, all forced transactions are included within one block tick (∼1 second) of their deadline expiry. The maximum observed latency is 1,010.6 ms. This is three orders of magnitude faster than the 24-hour deadline window, providing substantial safety margin. In practice, the forced inclusion mechanism drains the queue at every block production cycle, meaning forced transactions are never deferred beyond the first block after their deadline.

Finding 4: Concurrent access degrades insertion throughput but preserves correctness. Single-threaded mempool insertion achieves ∼2.8M tx/s, while concurrent access with 4 goroutines reduces this to ∼33K tx/s due to mutex contention. However, 33K tx/s still exceeds the target throughput (500 TPS) by a factor of 66×, and FIFO ordering remains perfect. For production use, the mutex-based approach is sufficient; lock-free alternatives (e.g., MPSC channels) are available as optimization options if contention becomes measurable.

Finding 5: The enterprise context eliminates MEV as a design concern. The zero-fee transaction model, absence of public DEX contracts, and single known enterprise operator collectively eliminate the economic incentives for MEV extraction. FIFO ordering provides fairness without the complexity of MEV-mitigation mechanisms (threshold encryption, commit-reveal schemes, fair ordering protocols) required in public L2 systems.

B. MEV Analysis

We analyze the three primary MEV attack vectors and their applicability to the Basis Network enterprise context:

The only MEV vector that applies in the enterprise context is transaction censoring, which is addressed by the forced inclusion mechanism. The remaining vectors require infrastructure (DEXs, lending protocols, fee markets) that does not exist on enterprise-operated chains.

C. Invariants Discovered

The following invariants are established by this investigation and must be preserved by downstream pipeline stages (Logicist, Architect, Prover):

1) **INV-SEQ-1 (FIFO Ordering)**: The sequencer SHALL order transactions strictly by arrival time within each block. No priority ordering, no fee-based reordering. Transactions from the forced inclusion queue are prepended before mempool transactions but maintain FIFO order within each source.
2) **INV-SEQ-2 (Forced Inclusion Liveness)**: Every transaction submitted to the forced inclusion queue on L1

TABLE XII
TICKER-BASED SCENARIO TEST RESULTS

Scenario	Blocks	TX Inserted	TX Included	TPS	Fill Ratio (%)	FIFO
Steady 100 TPS	6	325	325	49.9	10.8	100%
Steady 500 TPS	6	364	364	55.9	12.1	100%
Burst 1,000 TPS	4	226	226	49.9	11.3	100%
Max capacity	4	261	261	57.6	13.1	100%
Forced inclusion	6	341	361	55.3	12.0	100%
Adversarial censoring	6	342	362	55.4	12.1	100%

TABLE XIII
MEMPOOL PERFORMANCE SUMMARY

Metric	Value	Source
Insert (single-thread)	∼2.8M tx/s	Go bench
Insert (batch, direct)	2.3–8.6 K tx/s	Direct meas.
Insert (concurrent, 4P)	33 K tx/s	Concurrent test
Drain (500 tx batch)	1.3–1.8 K tx/s	Direct meas.
Memory per insert	168 bytes	Go bench
Capacity enforcement	100% (5/5 dropped)	Unit test

TABLE XIV
BENCHMARK VALIDATION AGAINST PUBLISHED DATA

Metric	Ours	Published	Verdict
Block prod. (100 tx)	0.019 ms	1–10 ms	Consistent
Mempool insert	365 ns/op	<1 ms	Consistent
FIFO accuracy	100%	100%	Exact match
Forced incl. latency	∼1 s	10m–24h	Consistent

SHALL be included in an L2 block within the configured deadline. The sequencer MUST drain the forced inclusion queue at the start of every block production cycle, before processing mempool transactions.

3) **INV-SEQ-3 (Block Production Regularity)**: The sequencer SHALL produce blocks at fixed intervals ($\Delta t = 1$ second). Block production MUST NOT be skipped when the mempool is empty—empty blocks maintain the chain's liveness guarantee and simplify downstream batch aggregation.
4) **INV-SEQ-4 (Mempool Capacity Bound)**: The mempool SHALL enforce a maximum capacity C_{max}. When full, new insertions are rejected with an explicit error (not silently dropped). This bounds memory consumption at $C_{max} \times 168$ bytes for the queue structure.
5) **INV-SEQ-5 (Single-Operator Trust Model)**: The sequencer operates under a trusted-but-verified model. The operator is trusted to include transactions fairly (FIFO), but the forced inclusion mechanism provides a cryptographic fallback: if the operator censors a transaction, the user can bypass the sequencer entirely via L1 submission. The ZK proof verifies the resulting state transition, not the ordering policy.

TABLE XV
MEV ATTACK VECTOR ANALYSIS

Attack	Prerequisite	Basis Status
Front-running	Fee priority, public mempool	Eliminated (FIFO)
Sandwich	DEX with slippage	No public DEXs
Liquidation	DeFi lending protocols	No DeFi protocols
Time-bandit	Block reorg incentive	Single operator
Censoring	Sequencer discretion	Forced inclusion

D. Forced Inclusion Contract Design

The L1 forced inclusion contract follows the Arbitrum DelayedInbox pattern:

Listing 2. Forced Inclusion L1 Contract Interface (Solidity-like pseudocode)

```
// User submits tx to L1 queue
function submitForcedTx(
  bytes calldata txData
) external {
  queue.push(ForcedTx({
    data: txData,
    sender: msg.sender,
    deadline: block.timestamp + DEADLINE,
    index: queue.length
  }));
}

// Anyone can force after deadline
function forceInclusion(
  uint256 index
) external {
  require(
    block.timestamp >= queue[index].deadline
  );
  // emit event for sequencer to process
}
```

The contract maintains a FIFO queue indexed by submission order. The sequencer monitors L1 events and includes pending forced transactions before their deadline. After the deadline, any user can trigger `forceInclusion()`, which emits an event that the sequencer must process in the next block.

E. Residual Risks

EVM execution time. Block production headroom (>999 ms) must accommodate EVM execution. At 500 TPS with 21K gas/tx simple transfers, Geth processes ∼10,000 TPS—providing a 20× safety margin. Complex contract interactions (higher gas per transaction) reduce this margin but remain well within budget for enterprise workloads.

L1 monitoring latency. The forced inclusion queue polls L1 for events. Avalanche's sub-second finality ensures that L1 events are visible within 1–2 seconds, but network latency and RPC polling intervals add variable delay. This latency

is bounded by the 24-hour deadline and does not affect correctness.

State growth. The mempool uses 168 bytes per transaction in the queue structure. At 500 TPS sustained, mempool memory grows at ∼84 KB/s. With a capacity limit of 100,000 transactions, peak memory is ∼16 MB—negligible for server hardware.

Windows timer resolution. Some direct measurements report 0 μs due to Windows timer granularity (∼15 ms). The Go native benchmarks resolve this by using the `testing.B` framework with nanosecond precision. Production benchmarks should be conducted on Linux for consistent timer behavior.

F. Limitations

No EVM execution. The prototype measures pure sequencer overhead (mempool drain + block assembly) without EVM transaction execution. This is by design: EVM execution is the domain of RU-L1 (BNR-2026-008). The combined system performance requires integration testing.

No network layer. Transaction arrival in the prototype is via direct function call, not RPC or P2P networking. Network latency and serialization overhead will add to end-to-end latency but do not affect sequencer-internal performance.

No persistent storage. Blocks are held in memory, not written to disk. WAL (Write-Ahead Log) and database persistence will add per-block overhead. The validium node's WAL (benchmarked in RU-V4) adds ∼0.1 ms per write, which is negligible.

Single-machine benchmarks. All measurements are on a single Windows 11 machine with 20 cores. Production deployment on Linux server hardware may show different absolute numbers but the relative relationships and scaling behavior are expected to hold.

VII. Conclusion

This paper investigated the design and performance of a single-operator sequencer for the Basis Network enterprise zkEVM L2. Through comparative analysis of eight production L2 systems, implementation of a Go prototype (∼500 LOC), and comprehensive benchmarks across scaling, concurrency, and adversarial scenarios, we establish the following conclusions:

1) Centralized sequencing with L1-based forced inclusion is the universal production pattern. All eight surveyed systems use centralized sequencers; forced inclusion via L1 provides the censorship resistance guarantee without sacrificing throughput.
2) Block production overhead is negligible relative to EVM execution. At 5,000 transactions per block, the sequencer consumes <1 ms of the 1-second block interval, leaving >99.9% of the time budget for EVM execution.
3) FIFO ordering is deterministic and requires no enforcement mechanism beyond atomic sequence numbering at insertion time. 100% ordering accuracy is maintained across all tested conditions, including concurrent access with 4 producer goroutines.
4) Forced inclusion achieves single-tick latency (∼1 second) under both cooperative and adversarial conditions, three orders of magnitude faster than the 24-hour deadline. 100% of forced transactions are included within the first block after deadline expiry.
5) The enterprise zero-fee model eliminates MEV as a design concern. FIFO ordering provides fairness without the complexity of MEV-mitigation mechanisms required in public L2 systems.
6) Five architectural invariants (INV-SEQ-1 through INV-SEQ-5) are established as constraints for downstream pipeline stages: the Logicist (TLA+ specification), the Architect (Go implementation), and the Prover (Coq verification).

A. Recommendations

For the Basis Network enterprise zkEVM L2 sequencer implementation:

1) Deploy a single-operator sequencer with 1-second block intervals. The enterprise operator runs the sequencer as part of their L2 node, with the forced inclusion contract on L1 providing the censorship resistance fallback.
2) Implement FIFO mempool with atomic sequence numbering. The mutex-based approach is sufficient for enterprise workloads (∼33K tx/s under contention, 66× above target).
3) Deploy an Arbitrum-style forced inclusion contract on the Basis Network L1 with a 24-hour deadline. The L1 contract maintains a FIFO queue; the sequencer monitors events and includes pending forced transactions at block production time.
4) Use hybrid batch strategy (size OR time trigger) as validated in the validium node research (RU-V4). Block production and batch aggregation are separate concerns: blocks are produced every 1 second, batches are submitted to L1 when size or time thresholds are met.
5) Do not implement MEV-mitigation mechanisms (threshold encryption, commit-reveal, fair ordering protocols). The enterprise zero-fee model makes these unnecessary and their complexity would degrade sequencer performance.

B. Future Work

Immediate next steps in the Basis Network R&D pipeline:

- **RU-L3 (State Management)**: Integrate the Poseidon-based sparse Merkle tree with the sequencer's block production loop, measuring the combined overhead of state updates during block execution.
- **Integration benchmarks**: Combine the EVM executor (RU-L1) with the sequencer to measure end-to-end block production latency including transaction execution.
- **Forced inclusion contract**: Implement and deploy the Solidity L1 contract for the forced inclusion queue, including gas cost measurements on the Basis Network L1.

- **Network layer**: Add gRPC/JSON-RPC transaction submission and measure end-to-end latency including serialization and network round-trip.
- **WAL integration**: Add Write-Ahead Log for block persistence and measure the overhead on block production latency.

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